

# FAIRNESS-AWARE RECIDIVISM CLASSIFICATION ON COMPAS

*Reproducing and diagnosing algorithmic bias*

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## THE QUESTION

# Where does the bias live?

### ProPublica, 2016

Of the defendants who never re-offended, Black defendants were labelled high-risk about twice as often as white ones.

### Northpointe (the vendor)

The tool is calibrated within groups. A score of 7 means the same thing whatever your race.

*Both sides are right. Once base rates differ between groups, calibration and equal false-positive rates are mathematically incompatible — the dispute can't be settled on its own terms.*

**So instead: if I build an ordinary classifier on the same data, and never show it COMPAS's scores, does it come out biased anyway?**

# A narrower, answerable question

- COMPAS is proprietary — its 137-item questionnaire and scoring formula are undisclosed.
- It's tempting to blame the bias on that specific, hidden formula.
- This project tests the alternative: that the disparity is a property of the data itself, not of any particular algorithm.
- Method: train a standard, off-the-shelf classifier on the same public data, never show it COMPAS's scores, and see whether it reproduces the bias anyway.

## Also tested:

Is removing race from the feature set — “fairness through unawareness” — enough to remove the bias?

## Answer, in short:

**No — and it fails for a specific, traceable reason.**

# ProPublica's COMPAS dataset

compas-scores-two-years.csv — 7,214 defendants screened in Broward County, Florida, 53 columns.

- ✓ **Screening window** Arrest and COMPAS screening must fall within 30 days of each other
- ✓ **Matched case** Records with no matching COMPAS case are removed
- ✓ **Charge type** Ordinary traffic offences are removed
- ✓ **Valid outcome** Records with a missing risk category are removed

**7,214 → 6,172**

defendants after cleaning · target two\_year\_recid is 45.5% positive — close to balanced, so accuracy is a meaningful headline metric and stratified splitting is appropriate

# 53 columns down to 5

*Not Filter, Wrapper, or Embedded selection — domain-knowledge elimination on three grounds:*

## Leakage

is\_recid, re-offence records, custody dates  
— recorded at or after the outcome being predicted

## Circularity

decile\_score is the object of the audit —  
using it as input would re-predict COMPAS  
from COMPAS

## Non-informative

names, IDs, dates, free-text charge  
descriptions carry no generalisable signal

**Kept — five screening-time attributes:**

age

priors\_count

sex

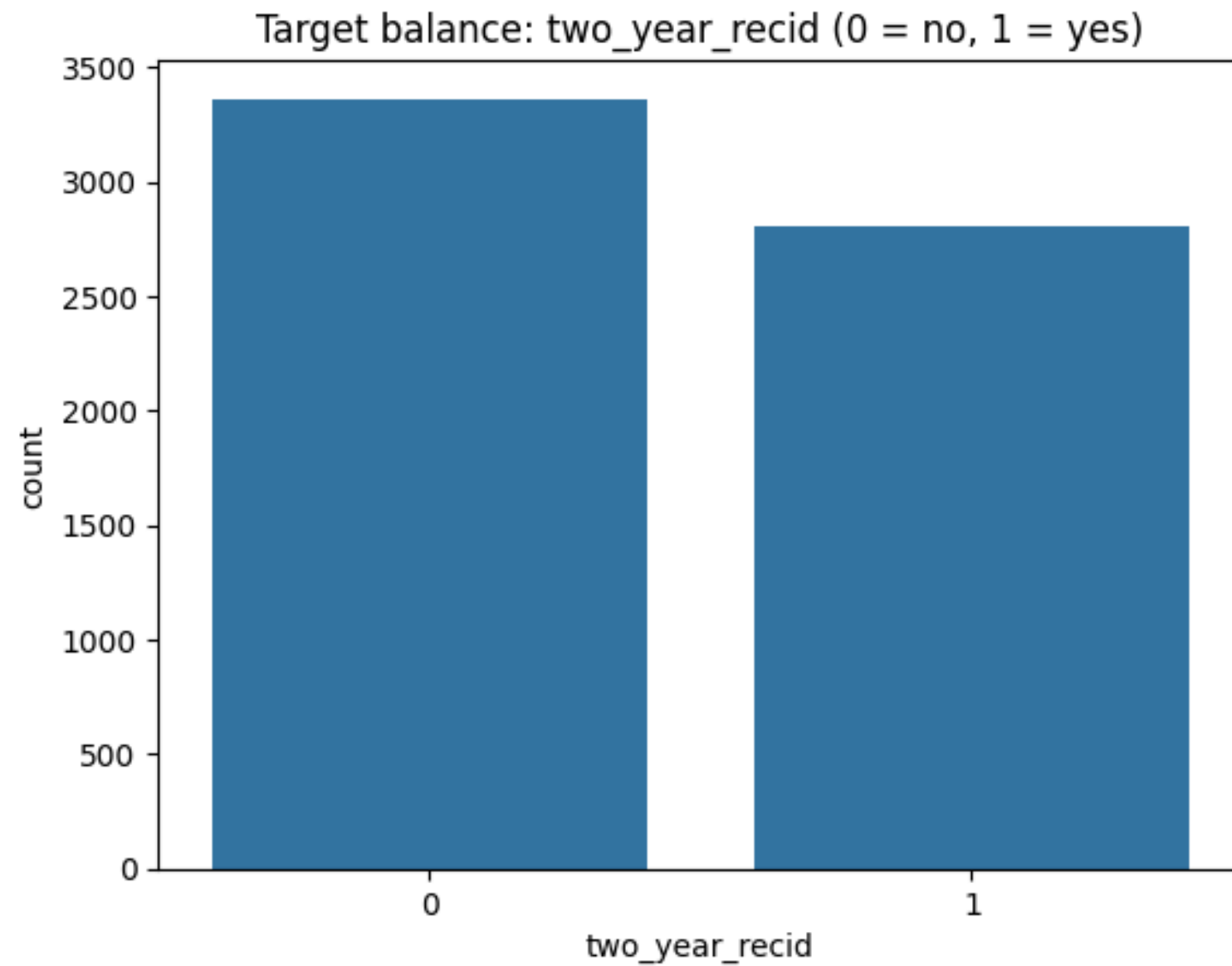
c\_charge\_degree

race

*race is kept in the baseline model as an ordinary one-hot predictor — deliberately, so it can be removed later to test whether that alone fixes the disparity (Section 8).*

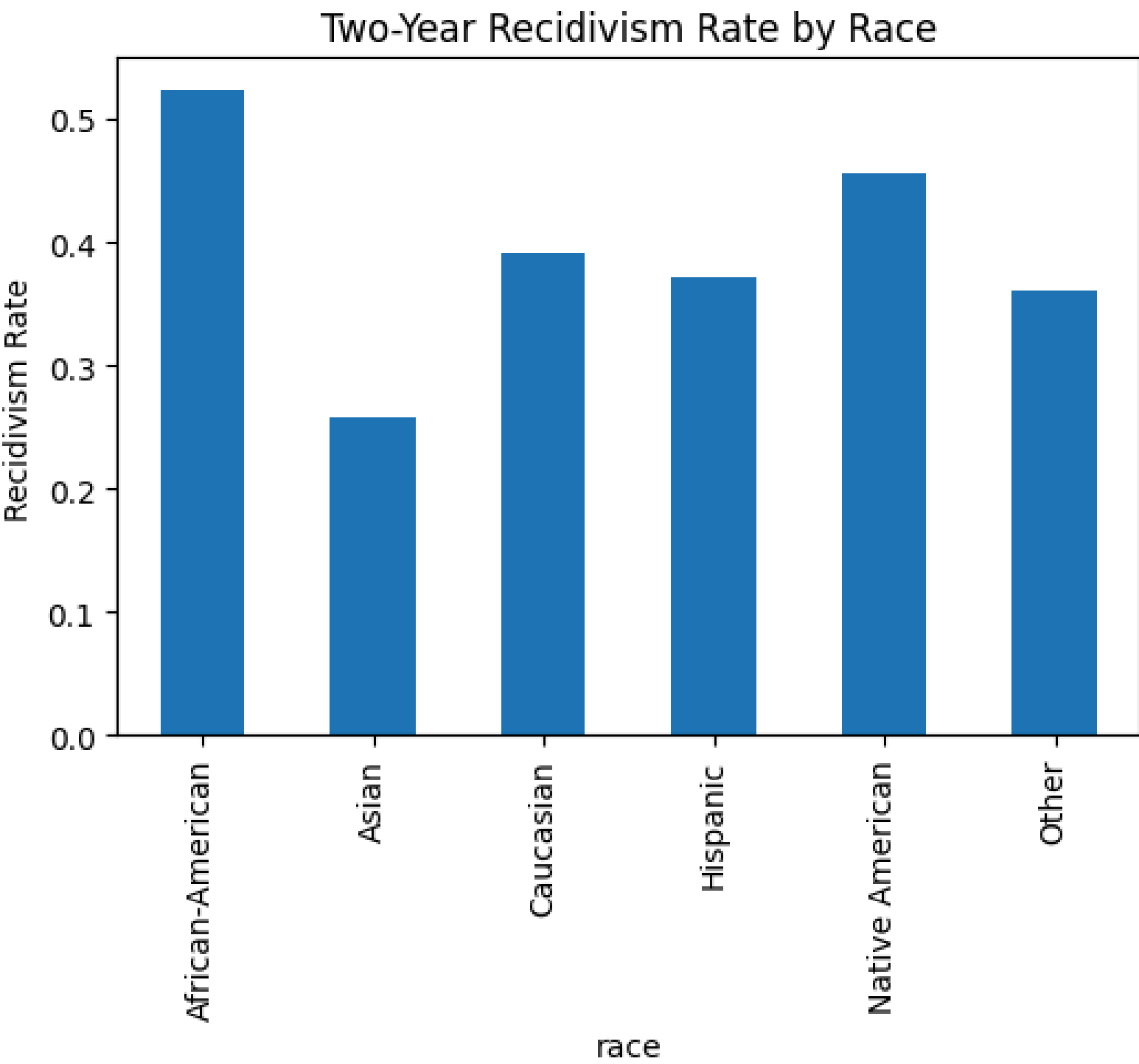
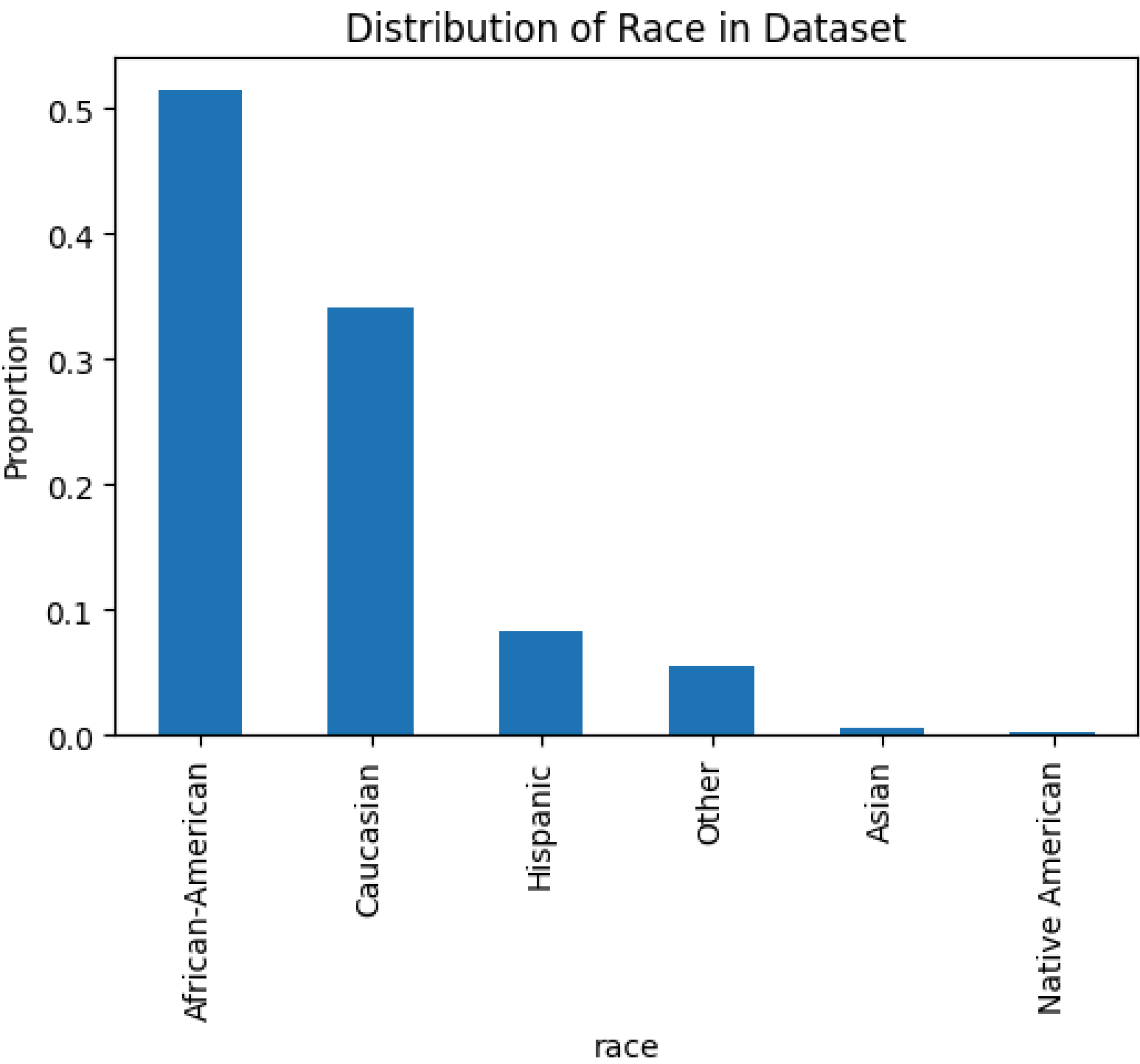


# Is the target balanced?

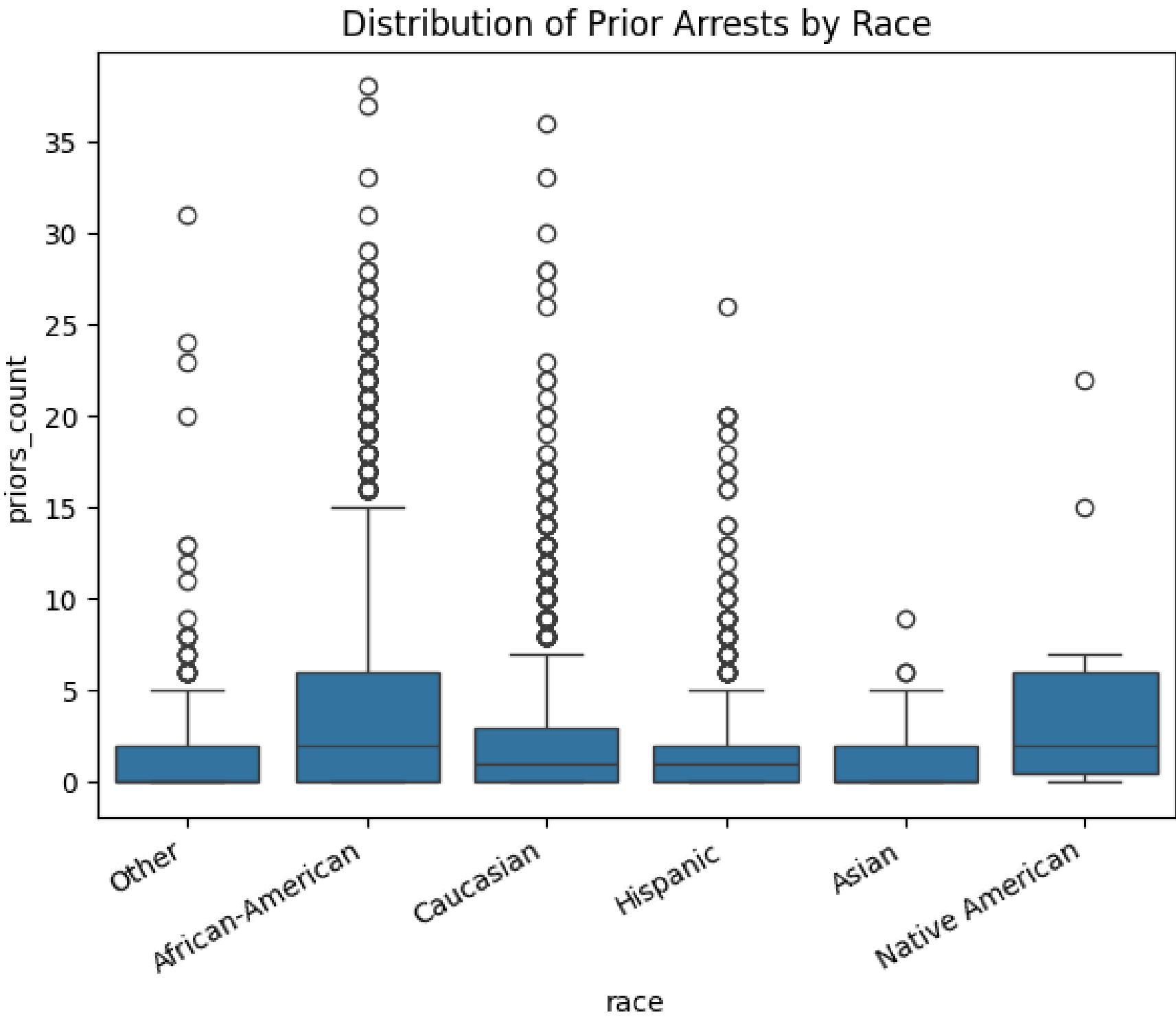


- two\_year\_recid: 45.5% positive / 54.5% negative
- Close to balanced → accuracy is a meaningful headline metric
- Motivates stratify=y on the train/test split, so both sets keep this proportion

# Sample composition and outcome by race



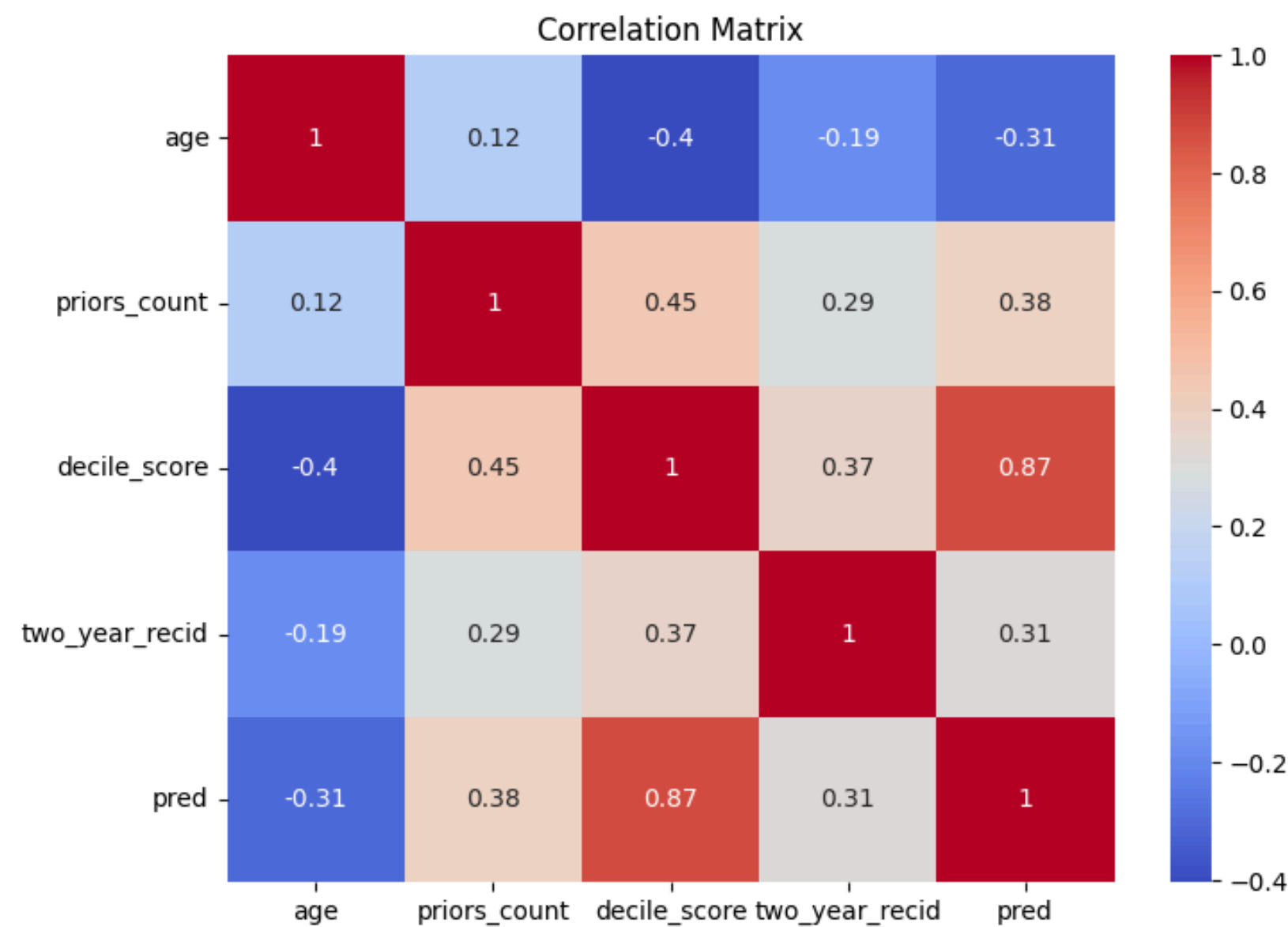
# Prior arrests, by race



- priors\_count is the strongest single predictor in this dataset
- Its distribution is visibly uneven across race
- First hint of the mechanism explored later: does this feature encode racial information?

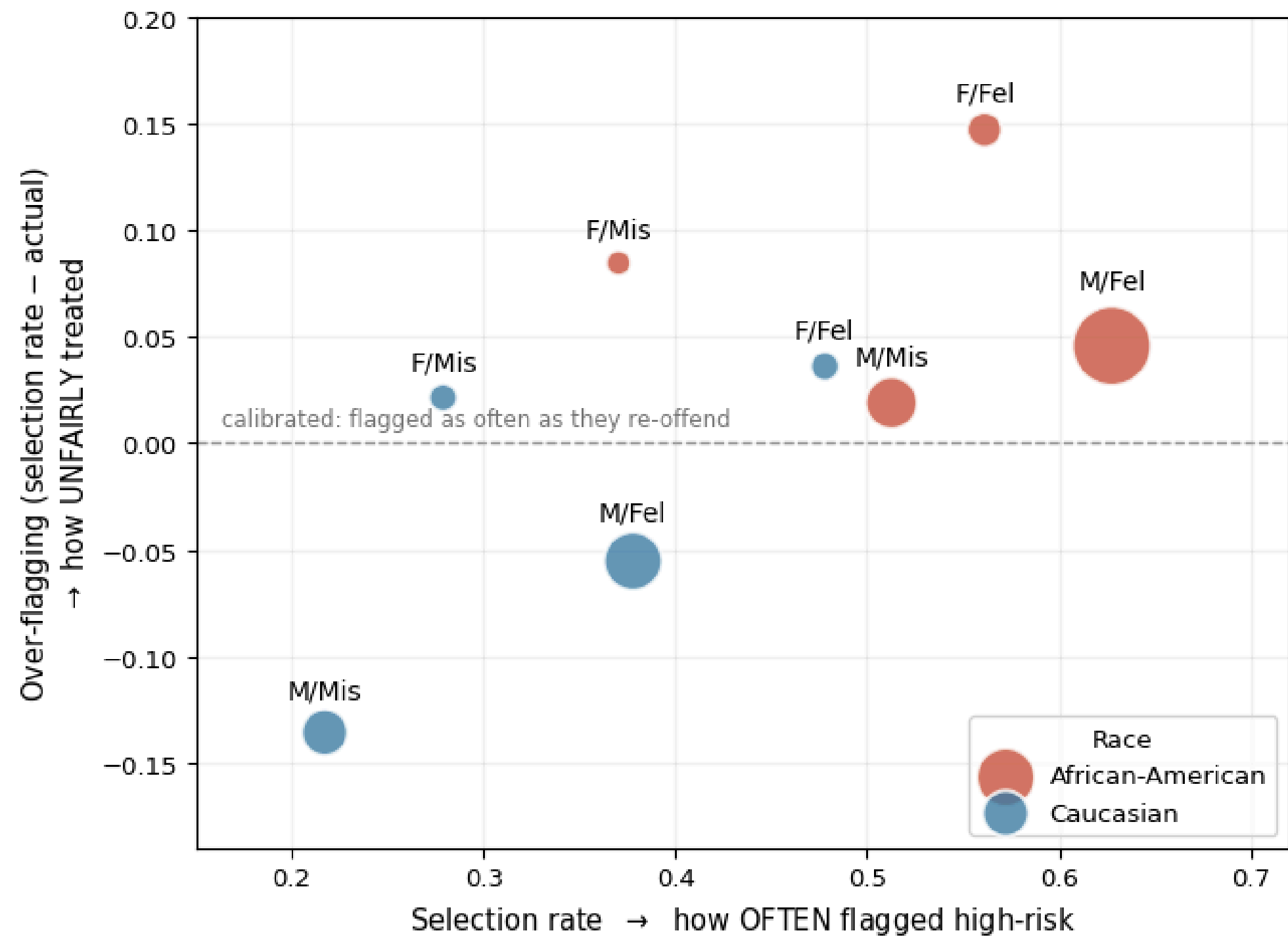


# Correlation, and COMPAS's own scores by race



*The correlation heatmap is descriptive only — it never drives a feature-selection decision here.*

# Flagged most often is not flagged most unjustly



- Horizontal axis: how OFTEN a group is flagged
- Vertical axis: how far that flag rate exceeds the group's real recidivism rate
- African-American male felony defendants are flagged most — the largest group
- But a different group is flagged most unfairly relative to their real outcome
- Aggregate metrics erase this distinction; only the disaggregated view reveals it

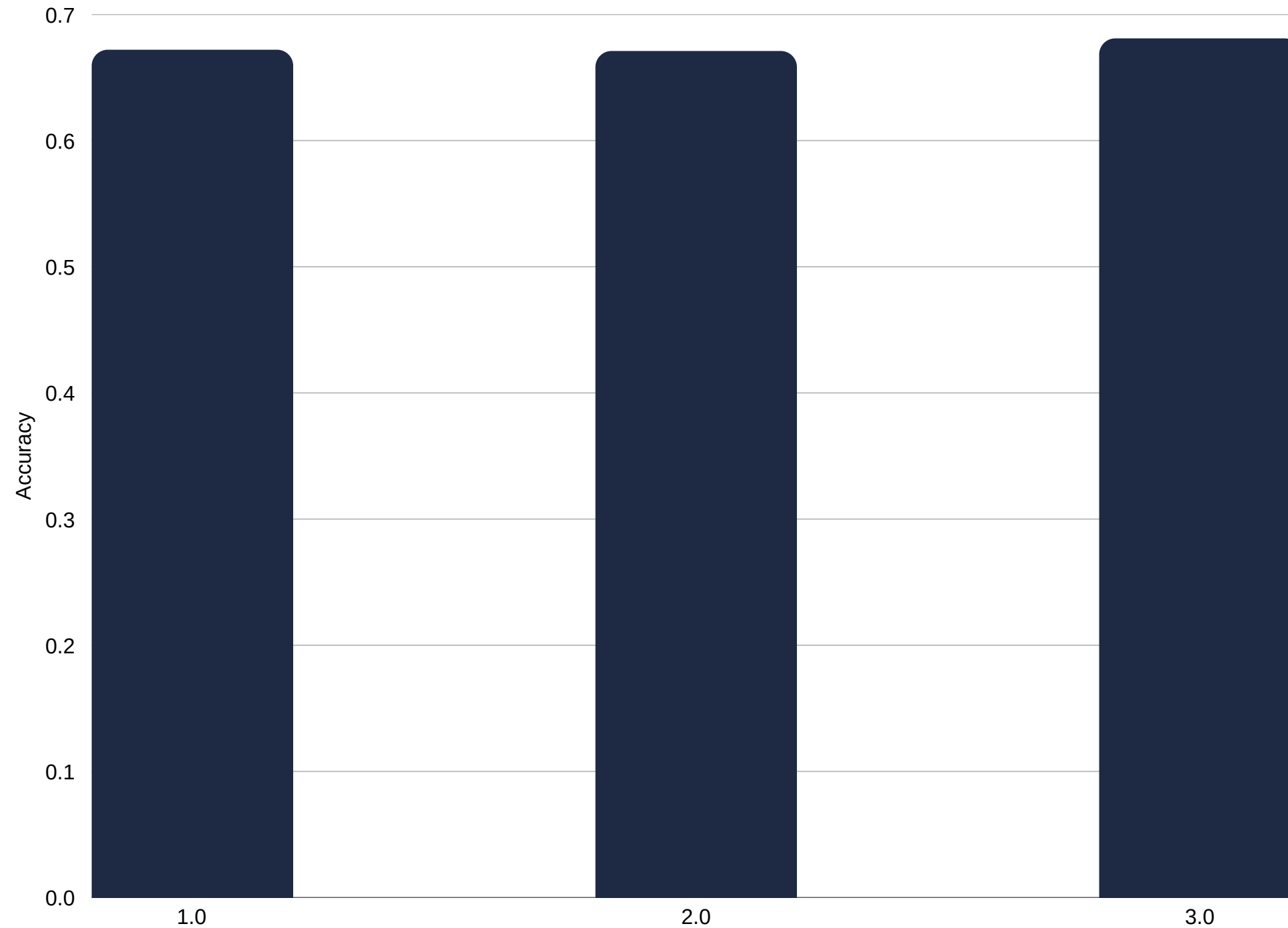
# Pipeline: preprocessing + estimator

- $X = [\text{race}, \text{sex}, \text{age}, \text{priors\_count}, \text{c\_charge\_degree}]$ ,  $y = \text{two\_year\_recid}$
- 70/30 train/test split, stratified on  $y$ , `random_state` fixed for reproducibility
- `ColumnTransformer`: numeric features pass through; categorical features one-hot encoded
- Wrapped in a single Pipeline with the classifier, so encoding is refit inside every cross-validation fold

## Why wrap it in a Pipeline?

Fitting the encoder once on the whole dataset before splitting would let test-set category information leak into training. Refitting inside every fold means no fold ever sees information from outside itself — the same leakage principle behind excluding `is_recid`, applied one level down at preprocessing.

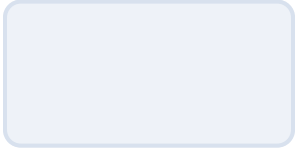
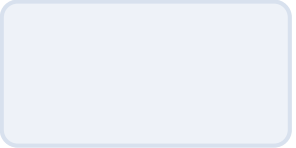
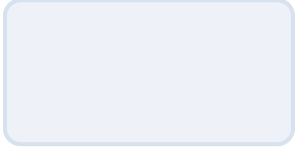
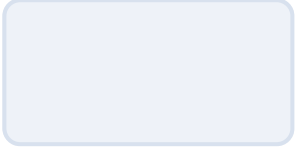
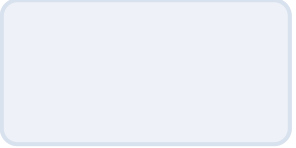
# Comparing model families, 5-fold cross-validation



- Trained on  $X_{\text{train}}$  only, 5 folds — test set never touched here
- All three cluster within one percentage point of each other
- Consistency across model families → the ceiling is imposed by the data, not model capacity
- Consistent with Dressel & Farid (2018): COMPAS-level accuracy is matched by a two-feature logistic regression

# Hyperparameter tuning: grid search

## Search space

|       | depth=4                                                                           | depth=6                                                                            | depth=8                                                                             |
|-------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| n=100 |  |  |  |
| n=200 |  |  |  |

*6 configurations, each scored by 5-fold CV — exhaustive, since the grid is small*

## Best parameters found

max\_depth = 6

n\_estimators = 100

**Test accuracy = 0.684 · AUC = 0.735**

## Why grid search, not random search?

Random search earns its keep on large search spaces — its whole pitch is coverage without the exhaustive cost. Here the space is only 2 hyperparameters × 6 combinations, so exhaustive grid search is already cheap: there's no efficiency gain left for random search to capture.

*cv=5 inside GridSearchCV — same fold-based scoring as the model comparison step, just repeated once per grid point.*



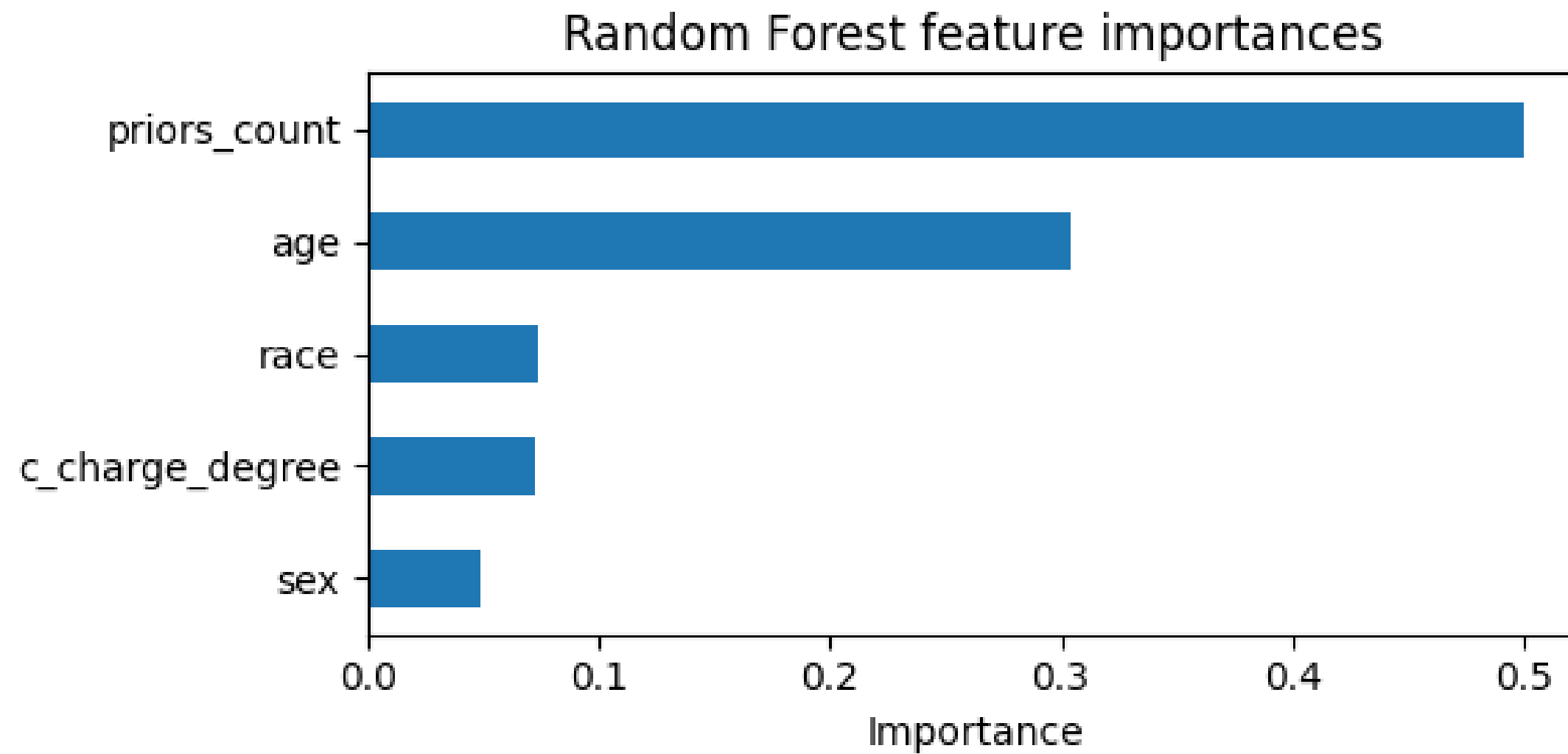
# Test-set accuracy in context

| Predictor                              | Accuracy |
|----------------------------------------|----------|
| Naive baseline (always majority class) | 0.545    |
| COMPAS decile score ( $\geq 5$ )       | 0.661    |
| Our tuned random forest                | 0.684    |

- +14 points over naive baseline
- Comparable to COMPAS — a gap of 41 defendants out of 1,852
- Achieved with 5 features vs. COMPAS's 137-item questionnaire
- Test set touched exactly once, after model selection was finished

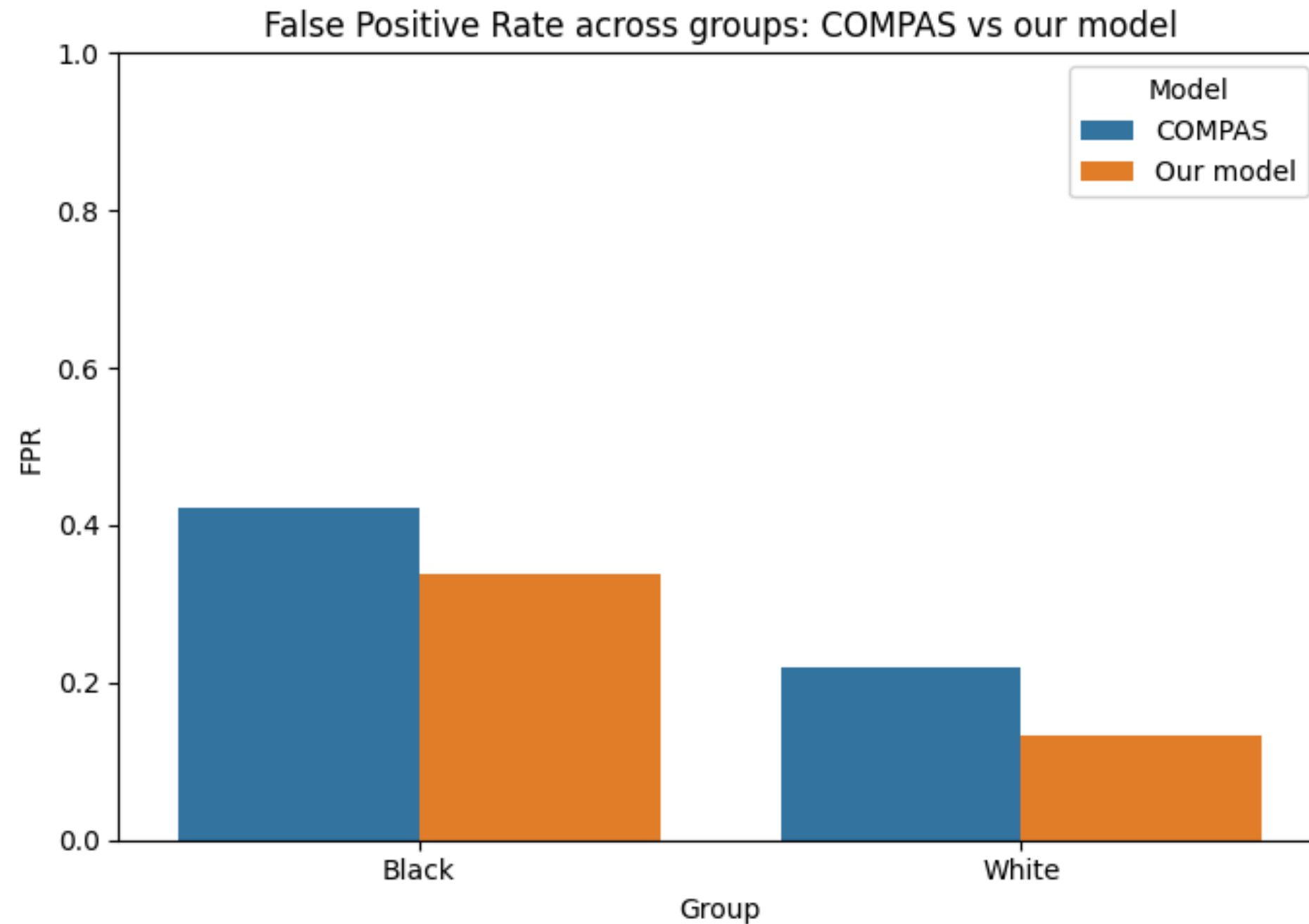
*AUC = 0.735 on the held-out test set*

## Feature importances — embedded, not selected on



- priors\_count dominates at 0.50, then age at 0.31
- race (0.07) and sex (0.05) trail well behind
- Reported as a diagnostic, not used to prune features — nothing was dropped based on this ranking
- The dominant feature is also the one most entangled with race

# Auditing our model — the naive comparison



- At the default 0.5 threshold, our model's FPR looks lower than COMPAS's for both groups
- Black defendants: 0.337 vs COMPAS's 0.411
- White defendants: 0.133 vs COMPAS's 0.227
- Read on its own, this looks like our model is the fairer predictor



# Why the appearance of fairness is an artefact

- FPR is not a property of a model — it's a property of wherever the threshold cuts the ranking
- A predictor that flags nobody has  $\text{FPR} = 0$  for every group — perfectly “fair” while being useless
- COMPAS flags 44.2% of the test set; our model at 0.5 flags only 37.3%
- Lower FPRs at 0.5 follow mechanically from flagging fewer people — they say nothing about bias

## The fix

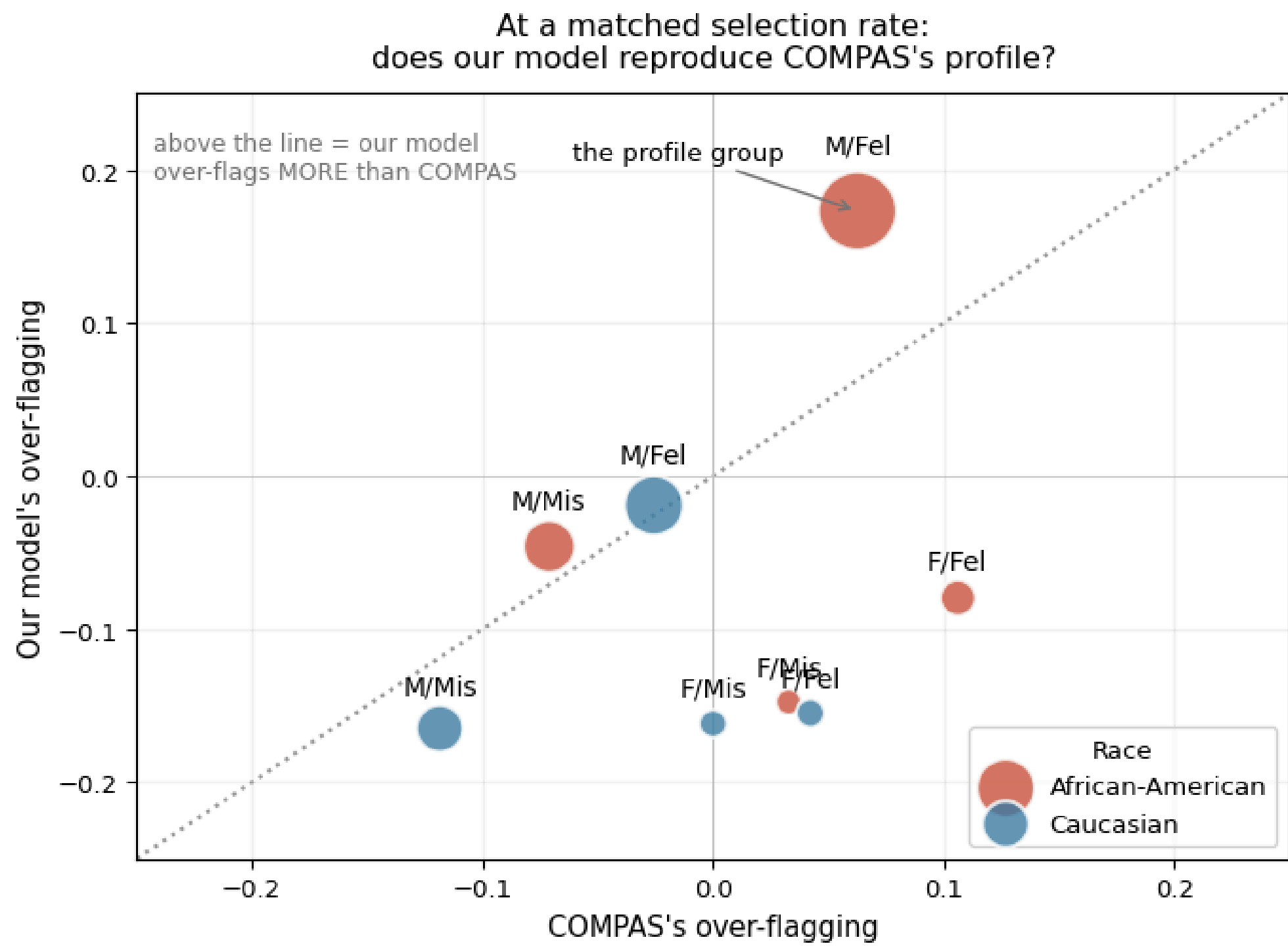
Lower our model's threshold to 0.461 — the point where it flags the same 44.2% of defendants COMPAS does. Compare fairness only at this matched operating point.

# The reversal, at a matched selection rate

|                | COMPAS (≥5) | Model (thr. 0.5) | Model (matched) |
|----------------|-------------|------------------|-----------------|
| Selection rate | 0.442       | 0.373            | 0.442           |
| Accuracy       | 0.661       | 0.684            | 0.684           |
| SPD            | 0.238       | 0.330            | 0.323           |
| FPR, Black     | 0.411       | 0.337            | 0.411           |
| FPR, white     | 0.227       | 0.133            | 0.188           |
| FPR ratio      | 1.81        | 2.53             | 2.19            |

At an equal willingness to flag, our model imposes the identical FPR on Black defendants as COMPAS, while treating white defendants more leniently — the disparity ratio rises from 1.81 to 2.19.

# The disparity is concentrated, not spread evenly



- Black men, felony charge (n=543, largest cell): COMPAS over-flags by 6.3 pts
- Our model over-flags the same group by 17.3 pts — nearly 3× COMPAS's excess
- White men, felony charge: essentially calibrated by our model (-1.9 pts)
- Remaining male cells sit near the identical-behaviour line — the disagreement is concentrated almost entirely on one cell

## MECHANISM

# priors\_count is a proxy for race

**51%**

of the sample is Black

**75%**

of the top decile of priors is Black

**16.8%**

of Black defendants fall in that decile (vs  
6.3% of white)

The top-priors decile re-offends at 75.7% vs. 41.6% for everyone else — strongly predictive, which is exactly what makes it dangerous as a feature.

*The model has a strong incentive to lean on the one feature that both predicts recidivism best and carries the most racial information — whether or not race itself is in the feature set.*

# Fairness through unawareness fails

Remove race, retrain with the same tuned hyperparameters, re-measure at the same threshold:

|            | With race | Without race | Change            |
|------------|-----------|--------------|-------------------|
| Accuracy   | 0.684     | 0.689        | ~unchanged        |
| SPD        | 0.330     | 0.272        | looks better      |
| FPR, Black | 0.337     | 0.339        | not helped at all |
| FPR, white | 0.133     | 0.173        | rises instead     |

The intervention does not reduce harm — it redistributes it. With priors\_count still present, the model reconstructs the racial signal it was denied.



# An introduced disparity: sex

Audited on the same terms, at the same matched operating point:

## COMPAS

Flags 45.6% of men, 38.3% of women

SPD = 0.072

FPR: 0.308 (men) vs 0.266 (women)

**Close to parity**

## Our model

Flags 49.7% of men, 21.4% of women

SPD = 0.283

FPR: 0.329 (men) vs 0.107 (women)

**≈ 4× COMPAS's disparity**

*Reported, not mitigated — a stated limitation of the project.*

## CONCLUSIONS

# The bias lives in the data, not the algorithm

### **Reproducible**

A conventional random forest, never shown COMPAS's scores, independently recovers the same directional bias — and amplifies it.

### **Traceable**

The mechanism is `priors_count` acting as a proxy for race: the strongest predictor available is also the most racially entangled feature.

### **Unawareness fails**

Removing race doesn't reduce harm for the disadvantaged group — it redistributes the cost onto the advantaged one.

### **Not automatically safer**

Building your own model with visible features and open code is not, by itself, a safeguard — it can introduce disparities of its own.

# Limitations

- priors\_count and charge records measure arrests, not offending — they reflect policing and enforcement patterns as much as behavior
- The target label inherits the same defect: re-arrest is only a proxy for re-offence, so every disparity here describes the criminal-justice record, not the people in it
- Intersectional cells shrink quickly — claims are restricted to cells with at least 50 defendants (20 in the test set)
- Only one mitigation strategy was tested. Reweighing (Kamiran & Calders) and equalised-odds post-processing (Hardt et al.) are the natural next steps, and the proxy analysis suggests they'd outperform unawareness





**The bias survives the removal of the attribute that names it.**

*Questions*